

Lesson 16 Smallest Counterexample induction + contradiction

Prop (same as Prop 1 in Lesson 14)

For all $n \in \mathbb{N}$,

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} \quad (*)$$

Proof by smallest counterexample

Suppose that the formula is not true for all $n \in \mathbb{N}$. Then there is a smallest counterexample $n = k$ for the formula $(*)$. We show that $k \neq 1$:

$$1 = \frac{1(1+1)}{2}$$

So $k > 1$, $k-1 > 0$, moreover $k-1$ is not a counterexample ($k-1 < k$, smallest counterexample)

Formula works for $k-1$:

$$1 + 2 + 3 + \dots + (k-1) = \frac{(k-1)k}{2}$$

But then, we can add k :

$$\begin{aligned} 1 + 2 + 3 + \dots + (k-1) + k &= \frac{(k-1)k}{2} + k \\ &= \frac{k}{2} ((k-1) + 2) = \frac{k}{2} (k+1) \end{aligned}$$

formula works for $n = k$!!!

This contradicts that k is a counterexample.

induction

$k \rightarrow k+1$
is true true

direct

Smallest counterex

k : smallest counterex

$k-1 \rightarrow k$
true true

contradiction

Fundamental Thm of Arithmetic
Every integer greater than 1 has
a unique prime factorization.

$$12 = \underline{2} \cdot \underline{2} \cdot \underline{3}$$

Proof of uniqueness

Let x be the smallest integer greater
1 which has 2 different prime
factⁿ's:

$$x = \underbrace{p_1 p_2 \cdots p_j}_{\text{primes}}, \quad x = \underbrace{q_1 q_2 \cdots q_l}_{\text{primes}}$$

x is not prime, because a prime has
a unique prime factⁿ, namely itself.

So, $j > 1$ and $l > 1$.

$$\text{Next, } p_1 | x \rightarrow p_1 | \underline{q_1 q_2 \cdots q_l}$$

$(p|ab \rightarrow pa \text{ or } pb)^{\text{prime}}$

$\rightarrow p_i | q_i \text{ for some } i$

$\rightarrow \underline{p_i} = \underline{q_i}$

$$x = \cancel{p_1} \underbrace{p_2 \cdots p_j}_{\text{prime}} = q_1 q_2 \cancel{q_i} q_l$$

$(p_i \neq 1)$ $\underline{x'} = p_2 \cdots p_j = q_1 q_2 \cdots \cancel{q_i} \cdots q_l$

$x' < x$ and x' has 2 different prime factorizations, contradicting that x is the smallest integer with 2 prime factorizations.